

Tutorial

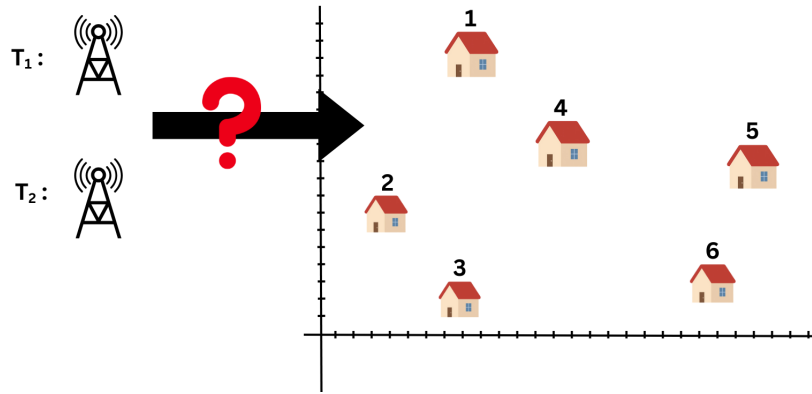
FML

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1. A telecom company wants to place $K = 2$ mobile towers T_1, T_2 as shown in below figure to serve 6 houses in a rural locality. The objective is to minimize the distance between each house and its nearest tower using K-means clustering.

House Coordinates:

House	X	Y
H1	2	2
H2	2	4
H3	5	8
H4	6	7
H5	9	2
H6	10	3



Tasks:

- (a) Apply K-means clustering with $K = 2$ to group the houses into two clusters. Choose initial tower locations as:

$$T_1 = H_1(2, 2), \quad T_2 = H_3(5, 8)$$

and determine which houses are served by Tower T_1 and Tower T_2 .

(b) If a new house $H_7(5, 3)$ is added, determine which tower will serve it.

2. Let us take a dataset of 11 points:

$(1, 2), (3, 4), (2.5, 4), (1.5, 2.5), (3, 5), (2.8, 4.5), (2.5, 4.5), (1.2, 2.5), (1, 3), (1, 5), (1, 2.5)$.

Apply DBSCAN to find the clusters for the given data (you can choose $\varepsilon = 0.6$, $\text{MinPts} = 4$).

3. Consider the following set of 6 one-dimensional data points:

18, 22, 25, 27, 42, 43

Apply the **Agglomerative Hierarchical Clustering** algorithm to build the hierarchical clustering dendrogram for the given data.

- (a) Use **Single Linkage** as the linkage criterion.
- (b) Use **Euclidean distance** as the distance metric.
- (c) At each step, merge the two clusters with the smallest single-linkage distance.
- (d) Update the proximity (distance) matrix after each merge.
- (e) Clearly show the proximity matrix corresponding to each iteration of the algorithm.
- (f) Continue the process until all points are merged into a single cluster.

Bootstrap Estimation of Standard Error

A team of analytics specialists is investigating the cycle time required to process loan applications. Based on prior experience, cycle time is assumed to be normally distributed with a mean of approximately 25 hours.

A recent random sample of $n = 10$ applications produced the following cycle times (in hours):

24.1514, 27.4145, 20.4000, 22.5151, 28.5152, 28.5611, 21.2489, 20.9983, 24.9840, 22.6245

The sample standard deviation of these observations is:

$$s = 3.11407$$

We wish to estimate the **bootstrap standard error** of the sample standard deviation.

Bootstrap Procedure

1. Generate $n_B = 200$ bootstrap samples, each of size $n = 10$, from a normal distribution with:

$$\mu = 25, \quad \sigma = 3.11417.$$

2. For each bootstrap sample, compute the sample standard deviation.

3. The first bootstrap sample is:

25.4274, 24.2272, 24.8565, 24.3458, 18.4343, 23.3179, 23.0699, 25.2876, 27.1541, 27.2932

4. The standard deviation of this first bootstrap sample is:

$$s_1 = 2.50635.$$

5. Repeat this process for all 200 bootstrap samples to obtain:

$$s_1, s_2, \dots, s_{200}.$$

6. Compute the mean of the bootstrap standard deviations:

$$\bar{s}_B = 3.03972.$$

7. The bootstrap estimate of the standard error is then:

$$SE_{\text{boot}} = 0.5464.$$

Conclusion

The bootstrap estimate of the standard error of the sample standard deviation is:

$$\boxed{0.5464}$$

This value is relatively large because the sample size ($n = 10$) is small, leading to greater variability in the estimate.

Method of Moments Estimation (Exponential Distribution)

Suppose $X_1, X_2, \dots, X_n$ is a random sample from an exponential distribution with parameter λ . The time to failure (in hours) of an electronic module in an automobile engine controller is assumed to be exponentially distributed. Eight units are randomly selected and tested, producing the following failure times:

$$11.96, 5.03, 67.40, 16.07, 31.50, 7.73, 11.10, 22.38.$$

Find the moment estimate of λ .

- (a) Using the **Method of Moments**, derive the estimator for λ given that for an exponential distribution:

$$E(X) = \frac{1}{\lambda}.$$

The sample mean is found to be $\bar{x} = 21.65$.

Maximum Likelihood Estimation (Exponential Distribution)

Suppose $X_1, X_2, \dots, X_n$ is a random sample from an exponential distribution with parameter λ .

- (a) Write the likelihood function for a random sample of size n from the exponential distribution:

$$f(x; \lambda) = \lambda e^{-\lambda x}, \quad x > 0.$$

- (b) Obtain the log-likelihood function and derive the maximum likelihood estimator (MLE) of λ .
- (c) The lifetimes (in hours) of $n = 6$ electronic components are observed as:

$$12.4, 5.6, 9.8, 15.2, 7.5, 10.5.$$

- (i) Compute the sample mean $\bar{x}$.
- (ii) Find the MLE of λ .

(a) Likelihood function:

$$L(\lambda) = \prod_{i=1}^n \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda \sum x_i}.$$

(b) Log-likelihood:

$$\ln L(\lambda) = n \ln \lambda - \lambda \sum x_i.$$

Differentiating and setting to zero:

$$\frac{d}{d\lambda} \ln L(\lambda) = \frac{n}{\lambda} - \sum x_i = 0.$$

Hence,

$$\hat{\lambda} = \frac{n}{\sum x_i} = \frac{1}{\bar{X}}.$$

(c)

$$\sum x_i = 61.0.$$

$$\bar{x} = \frac{61.0}{6} = 10.17.$$

$$\hat{\lambda} = \frac{1}{10.17} = 0.0983.$$